

实验二：进程间的协调通信

实验内容

在linux平台上实现1个父进程、10个以上子进程，其中一些子进程具有特定任务（如完成：显示、计算、报警等），各进程之间利用管道、信号灯和共享内存实现数据传送。

代码实现

```
#include <iostream>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/sem.h>
#include <cstring>
#include <ctime>
#include <vector>
#include <csignal>

#define NUM_CHILDREN 12 // 12个子进程
#define SHM_SIZE 1024 // 共享内存大小
#define SEM_KEY 1234 // 信号量键值
#define SHM_KEY 5678 // 共享内存键值

// 信号量操作联合体
union semun {
    int val;
    struct semid_ds *buf;
    unsigned short *array;
};

// 共享内存数据结构
struct SharedData {
    int counter;
    time_t last_update;
};

// 信号量P操作
void sem_wait(int semid) {
    struct sembuf sb = {0, -1, 0};
    semop(semid, &sb, 1);
}
```

```

// 信号量V操作
void sem_signal(int semid) {
    struct sembuf sb = {0, 1, 0};
    semop(semid, &sb, 1);
}

// 显示任务子进程
void display_task(int shmid, int semid) {
    SharedData* shared_data = (SharedData*)shmat(shmid, nullptr, 0);
    if (shared_data == (void*)-1) {
        perror("shmat");
        exit(1);
    }

    while (true) {
        sem_wait(semid);

        std::cout << "Display Process [" << getpid() << "]:\n";
        std::cout << "Counter: " << shared_data->counter << "\n";
        std::cout << "Last Update: " << ctime(&shared_data->last_update);
        std::cout << "-----\n";

        sem_signal(semid);
        sleep(3);
    }

    shmdt(shared_data);
}

// 计算任务子进程
void calculate_task(int shmid, int semid) {
    SharedData* shared_data = (SharedData*)shmat(shmid, nullptr, 0);
    if (shared_data == (void*)-1) {
        perror("shmat");
        exit(1);
    }

    while (true) {
        sem_wait(semid);

        shared_data->counter++;
        time(&shared_data->last_update);
        std::cout << "Calculate Process [" << getpid() << "]: Incremented
counter to "
                << shared_data->counter << "\n";

        sem_signal(semid);
        sleep(1);
    }
}

```

```

    shmdt(shared_data);
}

// 报警任务子进程
void alarm_task(int shmid, int semid) {
    SharedData* shared_data = (SharedData*)shmat(shmid, nullptr, 0);
    if (shared_data == (void*)-1) {
        perror("shmat");
        exit(1);
    }

    int last_counter = 0;
    while (true) {
        sem_wait(semid);

        if (shared_data->counter > last_counter + 5) {
            std::cerr << "ALERT Process [" << getpid() << "]: Counter
increased by more than 5! ("
                << last_counter << " -> " << shared_data->counter <<
                ")\n";
            last_counter = shared_data->counter;
        }

        sem_signal(semid);
        sleep(2);
    }

    shmdt(shared_data);
}

int main() {
    // 创建共享内存
    int shmid = shmget(SHM_KEY, sizeof(SharedData), IPC_CREAT | 0666);
    if (shmid == -1) {
        perror("shmget");
        exit(1);
    }

    // 初始化共享内存
    SharedData* shared_data = (SharedData*)shmat(shmid, nullptr, 0);
    if (shared_data == (void*)-1) {
        perror("shmat");
        exit(1);
    }

    shared_data->counter = 0;
    time(&shared_data->last_update);

    // 创建信号量
    int semid = semget(SEM_KEY, 1, IPC_CREAT | 0666);

```

```

if (semid == -1) {
    perror("semget");
    exit(1);
}

union semun su;
su.val = 1;
if (semctl(semid, 0, SETVAL, su) == -1) {
    perror("semctl");
    exit(1);
}

// 创建子进程
std::vector<pid_t> children;
for (int i = 0; i < NUM_CHILDREN; ++i) {
    pid_t pid = fork();
    if (pid == -1) {
        perror("fork");
        exit(1);
    } else if (pid == 0) {
        // 子进程
        switch (i % 3) {
            case 0:
                display_task(shmid, semid);
                break;
            case 1:
                calculate_task(shmid, semid);
                break;
            case 2:
                alarm_task(shmid, semid);
                break;
        }
        exit(0);
    } else {
        // 父进程记录子进程ID
        children.push_back(pid);
    }
}

sleep(20);

for (pid_t pid : children) {
    kill(pid, SIGTERM);
    waitpid(pid, nullptr, 0);
}

shmdt(shared_data);
shmctl(shmid, IPC_RMID, nullptr);
semctl(semid, 0, IPC_RMID);

```

```
std::cout << "Parent process exiting.\n";  
return 0;  
}
```

运行结果



hoshiuz@hoshiuz-virtual-machine: ~/lab2



```
hoshiuz@hoshiuz-virtual-machine:~/lab2$ ./process
```

```
Display Process [2593]:
```

```
Counter: 0
```

```
Last Update: Thu May 22 16:59:18 2025
```

```
-----
```

```
Calculate Process [2594]: Incremented counter to 1
```

```
Display Process [2596]:
```

```
Counter: 1
```

```
Last Update: Thu May 22 16:59:18 2025
```

```
-----
```

```
Calculate Process [2597]: Incremented counter to 2
```

```
Display Process [2599]:
```

```
Counter: 2
```

```
Last Update: Thu May 22 16:59:18 2025
```

```
-----
```

```
Calculate Process [2600]: Incremented counter to 3
```

```
Display Process [2602]:
```

```
Counter: 3
```

```
Last Update: Thu May 22 16:59:18 2025
```

```
-----
```

```
Calculate Process [2603]: Incremented counter to 4
```

```
Calculate Process [2594]: Incremented counter to 5
```

```
Calculate Process [2597]: Incremented counter to 6
```

```
Calculate Process [2600]: Incremented counter to 7
```

```
hoshiuz@hoshiuz-virtual-machine: ~/lab2
Calculate Process [2600]: Incremented counter to 7
Calculate Process [2603]: Incremented counter to 8
ALERT Process [2595]: Counter increased by more than 5! (0 -> 8)
Calculate Process [2594]: Incremented counter to 9
ALERT Process [2598]: Counter increased by more than 5! (0 -> 9)
Calculate Process [2600]: Incremented counter to 10
ALERT Process [2601]: Counter increased by more than 5! (0 -> 10)
Calculate Process [2597]: Incremented counter to 11
ALERT Process [2604]: Counter increased by more than 5! (0 -> 11)
Calculate Process [2603]: Incremented counter to 12
Calculate Process [2600]: Incremented counter to 13
Display Process [2599]:
Counter: 13
Last Update: Thu May 22 16:59:21 2025
-----
Calculate Process [2594]: Incremented counter to 14
Display Process [2602]:
Counter: 14
Last Update: Thu May 22 16:59:21 2025
-----
Calculate Process [2597]: Incremented counter to 15
Calculate Process [2603]: Incremented counter to 16
Display Process [2596]:
Counter: 16
```



Last Update: Thu May 22 16:59:21 2025

Display Process [2593]:

Counter: 16

Last Update: Thu May 22 16:59:21 2025

ALERT Process [2595]: Counter increased by more than 5! (8 -> 16)

ALERT Process [2598]: Counter increased by more than 5! (9 -> 16)

ALERT Process [2601]: Counter increased by more than 5! (10 -> 16)

Calculate Process [2600]: Incremented counter to 17

Calculate Process [2594]: Incremented counter to 18

Calculate Process [2597]: Incremented counter to 19

Calculate Process [2603]: Incremented counter to 20

Calculate Process [2600]: Incremented counter to 21

Calculate Process [2594]: Incremented counter to 22

Calculate Process [2597]: Incremented counter to 23

Calculate Process [2603]: Incremented counter to 24

ALERT Process [2595]: Counter increased by more than 5! (16 -> 24)

ALERT Process [2598]: Counter increased by more than 5! (16 -> 24)

ALERT Process [2601]: Counter increased by more than 5! (16 -> 24)

ALERT Process [2604]: Counter increased by more than 5! (11 -> 24)

Calculate Process [2600]: Incremented counter to 25

Calculate Process [2594]: Incremented counter to 26

Display Process [2599]:



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```
Counter: 26
Last Update: Thu May 22 16:59:24 2025
-----
Calculate Process [2597]: Incremented counter to 27
Display Process [2602]:
Counter: 27
Last Update: Thu May 22 16:59:24 2025
-----
Display Process [2593]:
Counter: 27
Last Update: Thu May 22 16:59:24 2025
-----
Display Process [2596]:
Counter: 27
Last Update: Thu May 22 16:59:24 2025
-----
Calculate Process [2603]: Incremented counter to 28
Calculate Process [2600]: Incremented counter to 29
Calculate Process [2594]: Incremented counter to 30
Calculate Process [2597]: Incremented counter to 31
Calculate Process [2603]: Incremented counter to 32
ALERT Process [2595]: Counter increased by more than 5! (24 -> 32)
ALERT Process [2598]: Counter increased by more than 5! (24 -> 32)
ALERT Process [2601]: Counter increased by more than 5! (24 -> 32)
```



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ALERT Process [2604]: Counter increased by more than 5! (24 -> 32)

Calculate Process [2600]: Incremented counter to 33

Calculate Process [2594]: Incremented counter to 34

Calculate Process [2597]: Incremented counter to 35

Calculate Process [2603]: Incremented counter to 36

Calculate Process [2600]: Incremented counter to 37

Calculate Process [2594]: Incremented counter to 38

Display Process [2599]:

Counter: 38

Last Update: Thu May 22 16:59:27 2025

Display Process [2593]:

Counter: 38

Last Update: Thu May 22 16:59:27 2025

Display Process [2602]:

Counter: 38

Last Update: Thu May 22 16:59:27 2025

Calculate Process [2597]: Incremented counter to 39

Display Process [2596]:

Counter: 39

Last Update: Thu May 22 16:59:27 2025

```
hoshiuz@hoshiuz-virtual-machine: ~/lab2
Calculate Process [2603]: Incremented counter to 40
ALERT Process [2595]: Counter increased by more than 5! (32 -> 40)
ALERT Process [2598]: Counter increased by more than 5! (32 -> 40)
ALERT Process [2601]: Counter increased by more than 5! (32 -> 40)
ALERT Process [2604]: Counter increased by more than 5! (32 -> 40)
Calculate Process [2597]: Incremented counter to 41
Calculate Process [2594]: Incremented counter to 42
Calculate Process [2600]: Incremented counter to 43
Calculate Process [2603]: Incremented counter to 44
Calculate Process [2597]: Incremented counter to 45
Calculate Process [2594]: Incremented counter to 46
Calculate Process [2600]: Incremented counter to 47
Calculate Process [2603]: Incremented counter to 48
ALERT Process [2595]: Counter increased by more than 5! (40 -> 48)
ALERT Process [2598]: Counter increased by more than 5! (40 -> 48)
ALERT Process [2601]: Counter increased by more than 5! (40 -> 48)
ALERT Process [2604]: Counter increased by more than 5! (40 -> 48)
Display Process [2599]:
Counter: 48
Last Update: Thu May 22 16:59:29 2025
-----
Display Process [2593]:
Counter: 48
Last Update: Thu May 22 16:59:29 2025
```



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Counter: 48

Last Update: Thu May 22 16:59:29 2025

Calculate Process [2597]: Incremented counter to 49

Calculate Process [2594]: Incremented counter to 50

Calculate Process [2600]: Incremented counter to 51

Calculate Process [2603]: Incremented counter to 52

Calculate Process [2597]: Incremented counter to 53

Calculate Process [2594]: Incremented counter to 54

Calculate Process [2600]: Incremented counter to 55

Calculate Process [2603]: Incremented counter to 56

ALERT Process [2595]: Counter increased by more than 5! (48 -> 56)

ALERT Process [2598]: Counter increased by more than 5! (48 -> 56)

ALERT Process [2601]: Counter increased by more than 5! (48 -> 56)

ALERT Process [2604]: Counter increased by more than 5! (48 -> 56)

Calculate Process [2597]: Incremented counter to 57

Calculate Process [2594]: Incremented counter to 58

Calculate Process [2603]: Incremented counter to 59

Calculate Process [2600]: Incremented counter to 60

Display Process [2599]:

Counter: 60

Last Update: Thu May 22 16:59:32 2025

Display Process [2602]:



```
-----  
Display Process [2593]:  
Counter: 72  
Last Update: Thu May 22 16:59:35 2025  
-----  
Display Process [2596]:  
Counter: 72  
Last Update: Thu May 22 16:59:35 2025  
-----  
Display Process [2602]:  
Counter: 72  
Last Update: Thu May 22 16:59:35 2025  
-----  
ALERT Process [2595]: Counter increased by more than 5! (66 -> 72)  
Calculate Process [2594]: Incremented counter to 73  
Calculate Process [2600]: Incremented counter to 74  
Calculate Process [2603]: Incremented counter to 75  
ALERT Process [2598]: Counter increased by more than 5! (67 -> 75)  
ALERT Process [2604]: Counter increased by more than 5! (67 -> 75)  
ALERT Process [2601]: Counter increased by more than 5! (68 -> 75)  
Calculate Process [2597]: Incremented counter to 76  
Calculate Process [2594]: Incremented counter to 77  
Calculate Process [2600]: Incremented counter to 78  
Calculate Process [2603]: Incremented counter to 79
```